

Control work 3

• Coordinates, vectors and area

The Quarter III assessment, and the four errors it produces every year.

LESSON 52	1 × 40 MIN	GEOMETRY 8, ТЕМЫ 53–54	STAGE 9 · 7.1–7.2, 13.1–13.2
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LESSON CLOCK

2'

36'

2'

Setting up2 min

The paper36 min

Collect in2 min

LEARNING OBJECTIVES

• Assess coordinates, distance, vectors and all the area formulas.

• Diagnose each lost mark by error type.

• Re-solve every task that was lost.

TEXTBOOK

NationalТемы 53–54, pp. 126–127

CambridgeLearner’s Book pp. 142–155, 284–296

WorkbookWorkbook 7.1–7.2

01 Explanation

The paper (40 minutes)

Two variants of eight tasks. Tasks 1–5 are worth 1 mark, tasks 6–8 are worth 2:

- task 1 · the midpoint of a segment
- task 2 · the distance between two points
- task 3 · the equation of a circle from centre and radius
- task 4 · AB and $|AB|$ from two points
- task 5 · the area of a parallelogram or a triangle
- task 6 · the area of a trapezium or a rhombus
- task 7 · a compound shape
- task 8 · a coordinate proof

Marking note: the four errors below account for almost every lost mark in this paper.

The four errors

WORK ON MISTAKES

- Start minus end** in a vector — the answer comes out with both signs reversed.
- The slant side used as the height** of a parallelogram or trapezium.

3. **The trapezium formula without the halving** — an answer exactly twice too big.

4. **Radius and diameter swapped**, or r^2 read off as r in the circle equation.

Each pupil writes the error number beside every lost mark and re-solves that task in full.

02 Worked examples

EXAMPLE 1 Find the error: for $A(1; 2)$ and $B(5; 7)$, $AB = (-4; -5)$

① The subtraction went the wrong way round.
 AB means “from A to B ”.

② $AB = (5 - 1; 7 - 2)$
 End minus start.

③ $AB = (4; 5)$
 The answer given is BA .

ANSWER

$(4; 5)$

EXAMPLE 2 Find the error: a trapezium with parallel sides 6 and 10 and height 4 has area 64

- ① The halving was left out.
 $(6 + 10) \cdot 4 = 64$ is the parallelogram, not the trapezium.

② $S = \frac{6 + 10}{2} \cdot 4$

③ $= 8 \cdot 4 = 32$
Check: less than $10 \cdot 4 = 40$ ✓

ANSWER

32

03 Interactive model

Show the wrong working, take a vote on the error type, then reveal.

Diagnose the error

Claimed: for $A(3; 1)$, $B(8; 4)$, $AB = (-5; -3)$

① AB

runs **from A to B**

· The subtraction is end minus start.

② $(8 - 3; 4 - 1)$

③ $= (5; 3)$

What
was
written
is
 BA .

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Control work	Nazorat ishi	Контрольная работа
Work on mistakes	Xatolar ustida ishlash	Работа над ошибками
Midpoint	O'rta nuqta	Середина
Distance	Masofa	Расстояние
Vector	Vektor	Вектор
Height	Balandlik	Высота
Trapezium	Trapetsiya	Трапеция
Units of area	Yuza birliklari	Единицы площади

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

For $A(2; 5)$ and $B(7; 9)$, AB is:

$(-5; -4)$

$(5; 4)$

$(9; 14)$

$(4; 5)$

A trapezium with parallel sides 5 and 9 and height 6 has area:

84

42

27

70

The radius of $(x - 1)^2 + (y - 2)^2 = 36$ is:

36

6

18

12

A parallelogram with sides 7 and 4 and an angle of 30° has area:

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. **Variant 1, task 1.** Find the midpoint of $(1; 3)$ and $(7; 9)$.

ANSWER $(4; 6)$

2. **Task 2.** Find the distance between $(0; 0)$ and $(5; 12)$.

ANSWER 13

3. **Task 3.** Write the equation of the circle, centre $(2; -3)$, radius 5.

ANSWER $(x - 2)^2 + (y + 3)^2 = 25$

4. **Task 4.** Find AB and $|AB|$ for $A(1; 1)$, $B(4; 5)$.

ANSWER $(3; 4), 5$

5. **Task 5.** Find the area of a triangle with base 12 and height 5.

ANSWER 30

6. **Task 6.** Find the area of a trapezium with parallel sides 6 and 10 and height 5.

ANSWER 40

7. **Task 7.** An L-shape is a 9×6 rectangle less a 3×2 corner. Find its area.

ANSWER 48

Medium · the standard question

7 PROBLEMS

1. **Variant 2, task 1.** Find the midpoint of $(-4; 6)$ and $(2; -2)$.

ANSWER $(-1; 2)$

2. **Task 2.** Find the distance between $(-1; 2)$ and $(3; 5)$.

ANSWER 5

3. **Task 3.** State the centre and radius of $(x + 4)^2 + (y - 1)^2 = 49$.

ANSWER $(-4; 1), r = 7$

4. **Task 4.** Find AB and $|AB|$ for $A(-2; 3), B(4; 11)$.

ANSWER $(6; 8), 10$

5. **Task 5.** Find the area of a parallelogram with base 11 and height 6.

ANSWER 66

6. **Task 6.** Find the area of a rhombus with diagonals 10 and 12.

ANSWER 60

7. **Task 8.** Show that $(0;0), (5;0), (5;4), (0;4)$ is a rectangle.

ANSWER Both diagonals have midpoint $(2.5; 2)$ and both equal $\sqrt{41}$.

Hard · stretch and reason

7 PROBLEMS

1. Find the fourth vertex of a parallelogram with $(1;2)$, $(6;3)$, $(8;8)$.

ANSWER $(3; 7)$

2. A rhombus has perimeter 40 and one diagonal 16. Find its area.

ANSWER side 10, other diagonal 12, area 96

3. A trapezium has area 90, height 9 and one parallel side 8. Find the other.

ANSWER 12

4. Find the area of the polygon $(0;0)$, $(8;0)$, $(8;5)$, $(4;8)$, $(0;5)$.

ANSWER 52

5. $a = (3; -2)$ and $b = (-1; 5)$. Find $2a - b$ and its length.

ANSWER $(7; -9)$, $\sqrt{130}$

6. A circle has centre $(2; 3)$ and passes through $(6; 6)$. Find its equation and its area.

ANSWER $(x - 2)^2 + (y - 3)^2 = 25$, area 25π

7. Show that $(1;1)$, $(5;3)$, $(7;7)$, $(3;5)$ is a parallelogram, and find its area.

ANSWER Diagonals share the midpoint $(4; 4)$; area 12

07 Homework

Homework — 5 problems

Re-solve every task you lost marks on, with the error number written beside it.

1. Re-solve, in full, each task you lost a mark on.
2. Beside each, write the error number (1–4) that caused it.
3. Find AB and $|AB|$ for $A(3; -1)$, $B(9; 7)$.
4. Find the area of a trapezium with parallel sides 9 and 15 and height 8.
5. State the centre and radius of $(x - 5)^2 + (y + 2)^2 = 64$.